

# Tidal: Direction-Paired Virtual-Channel Buffer Pooling with a 1R1W Shared Pool

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Equal contribution: 50% each

## Abstract

Conventional on-chip routers bind virtual-channel (VC) storage to individual input directions. This organization strands storage when one direction is congested and its opposite peer is idle. TIDAL reorganizes the same eight VC slots per opposing-direction pair into two two-slot reserved regions and a four-slot shared POOL. Each direction retains one private adaptive VC and one private escape VC. The POOL provides only one read and one write service per cycle; alternating read phases let one side inspect its reserved region while the other inspects the POOL. Four conserved POOL credits control admission. A pressure policy moves released credits toward the side with fuller reserved storage, keeps the previous owner on ties, caps ownership at three, and reclaims only idle peer-owned slots on demand.

We implement the design in gem5 Garnet and evaluate an equal-storage four-private-VC baseline. Across 651 main-matrix simulations, all runs completed without deadlock and recorded zero cross-direction POOL read conflicts. Under persistent 90/10 X-direction skew with eight-cycle links, TIDAL lowers network latency by 9.2% and extends the coarse stable offered rate from 0.08 to 0.10. At a fully skewed point beyond baseline saturation it delivers 1.818 times as many packets. Under tornado traffic, pressure-aware credit control lowers latency by 4.85%, whereas blind round-robin ownership loses 56.5% throughput. Balanced, short-link, and rapidly reversing traffic can be slower. The results identify persistent directional skew near an input credit limit as the useful operating region and motivate timing and RTL cost validation.

**Keywords:** on-chip networks, virtual channels, shared buffers, credit flow control, Garnet, gem5

## 1 Introduction

Virtual-channel flow control improves throughput by multiplexing logical channels over physical links, but it also makes buffer organization a central router design choice [3]. Garnet models a pipelined, credit-controlled network inside gem5 and is widely used to study these tradeoffs [1, 2]. Its conventional private organization is simple: input-direction VCs cannot migrate. When traffic is directionally skewed, a busy side can therefore stall while unused capacity remains stranded at its opposite peer.

Fully shared buffers improve utilization, but unconstrained sharing requires multiported storage or arbitration that can erase the intended cost benefit. Our question is narrower: *can an equal-capacity, direction-paired buffer recover useful elasticity while exposing only one shared read and one shared write service per pair and cycle?*

This paper makes four contributions:

- an equal-storage organization with three independent pairs per router, one private adaptive VC and one escape VC per direction, and four movable adaptive slots per pair;
- an alternating read schedule and a one-write admission guard that implement the shared bank as 1R1W;
- a conserved ownership-credit policy combining pressure, sticky ties, a three-credit cap, and idle-slot reclaim; and
- a 651-run evaluation that reports both the favorable region and the workloads where the fixed phase schedule is counterproductive.

## 2 Background and Motivation

A conventional four-VC input assigns four physical VC slots to each direction. Credit round-trip time determines how quickly a released downstream slot can be reused. With one-flit control traffic and an approximately 17-cycle credit loop at link latency  $L = 8$ , four slots impose an idealized zero-contention input-credit bound of

$$\lambda_{\text{credit}} \approx \frac{4}{17} = 0.235 \text{ flits/cycle.} \quad (1)$$

The exact saturation point can be lower because routing, output contention, and switch arbitration intervene.

TIDAL targets the case in which one member of an opposing pair repeatedly approaches this limit while the peer is underused. It does not attempt to replace the escape channel used by deadlock-free routing. Deterministic escape resources remain private, consistent with the separation of deadlock avoidance from adaptive resource use [4, 5].

## 3 Architecture

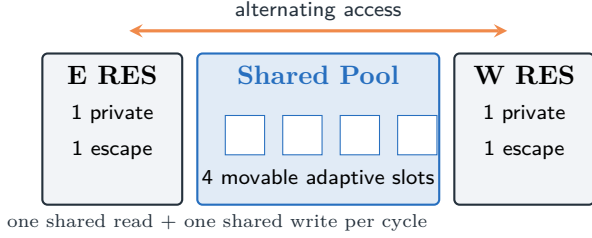
### 3.1 Direction-paired storage

Each router instantiates East/West, North/South, and Up/Down pairs. Figure 1 shows one pair. For each direction, RES contains one private adaptive VC and one private escape VC. The pair shares four adaptive slots:

$$2(1_{\text{private}} + 1_{\text{escape}}) + 4_{\text{pool}} = 8 \text{ slots.} \quad (2)$$

This equals the eight slots in two four-VC private inputs, so performance comparisons preserve total storage.

POOL membership follows *true ingress*, not the selected output. A flit entering from X may occupy the X-pair



**Figure 1:** One direction pair. Every router contains three copies: E/W, N/S, and U/D. Capacity matches the private four-VC baseline.

POOL and later depart through Y or Z. Each pooled flit stores its ingress direction and upstream VC so that its credit returns to the correct sender.

### 3.2 One-read, one-write service

The POOL accepts at most one read and one write service per pair, control virtual network, and cycle. Read phases are complementary:

| Phase | Side 0 (E/N/U) | Side 1 (W/S/D) |
|-------|----------------|----------------|
| Even  | RES            | POOL           |
| Odd   | POOL           | RES            |

Thus E can inspect RES while W inspects the POOL, and the roles reverse next cycle. Cycle parity structurally eliminates simultaneous cross-direction POOL reads. Ordinary VC and switch arbitration still select among eligible flits inside the active bank.

A per-pair write guard permits one adaptive POOL allocation each cycle. If both sides arrive, the second arrival may still use its private adaptive VC when that reserved credit is available; otherwise it waits. Escape storage never enters the shared bank and follows deterministic non-wrapping XYZ routing.

## 4 Pool-Credit Control

Each of the four shared slots has exactly one logical owner. For the E/W pair, ownership obeys

$$C_E + C_W = 4, \quad 1 \leq C_E, C_W \leq 3. \quad (3)$$

An owned POOL credit authorizes the corresponding upstream direction to reserve an idle slot. Private and escape credits remain pinned.

**Pressure grant.** When a pooled flit departs, its released credit is assigned to the side whose RES occupancy is larger. This local pressure signal approximates near-term need without adding a traffic predictor.

**Sticky tie.** Equal pressure preserves the previous owner. Stickiness avoids credit oscillation and favors the side likely to continue a burst.

**Owner cap.** A cap of three lets a hot side borrow one additional slot while guaranteeing one shared credit to its peer. Cap four is evaluated only as a full-elasticity upper bound.

**On-demand reclaim.** Release-only movement can freeze if an idle owner has no flit to release. A requester

below its cap can therefore take an *idle* peer-owned slot immediately. Busy slots are never revoked. In the current simulator, this owner-table update adds no cycle; Section 8 treats this as an optimistic timing assumption.

Blind owner round-robin is a useful control policy, but it is distinct from the round-robin arbitration inside the read scheduler. Rotating ownership without observing demand may park capacity at an inactive side.

## 5 Garnet Implementation

We implemented TIDAL in gem5 Garnet’s router, input unit, VC allocator, and switch-allocation paths. The exact strict-1R1W snapshot is retained at 1R1W/overlay; its provenance records implementation commit dc43a290 and base commit 2eb6a4e. Major additions are:

- physical-to-logical mapping for pair-global POOL slots;
- true-ingress and upstream-VC metadata for routing and credit return;
- complementary RES/POOL eligibility in switch allocation;
- a one-write-per-pair guard in downstream allocation;
- owner tables, pressure grants, cap enforcement, sticky ties, and idle-slot reclaim; and
- structural counters and X-biased/reversal synthetic traffic.

Experiments inject only virtual network 0 with one-flit control packets and one slot per control VC. Hence  $L = 8$  means an eight-cycle link, not an eight-flit packet. Exported counters include POOL allocations, owned allocations, demand reclaims, release handoffs/keeps, owner migrations, write blocks, bank reads, per-direction activity, and peak borrowing.

## 6 Evaluation

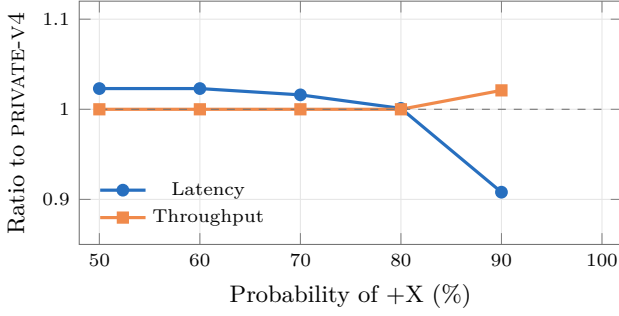
### 6.1 Methodology

The primary baseline, PRIVATE-V4, dedicates four physical VCs to every input. All TIDAL arms use two RES slots per side plus four shared slots per pair, preserving eight slots. We compare a non-migrating round-trip-static policy (RTS), blind owner round-robin, pressure cap three, and pressure cap four.

Tests use  $4 \times 4 \times 4$  or  $8 \times 4 \times 4$  3D torus networks, link latencies  $L \in \{1, 8\}$ , one-flit control packets, 10,000 simulated cycles for main long runs, and three seeds per reported point. The 651-run main matrix contains 120 broad validation, 96 policy, 72 owner-cap, 192 reversal, and 171 X-biased/VC-scaling runs. A separate 48-run recovery-instrumentation rerun is not included in this total.

### 6.2 Structural validation

All 651 main runs completed without deadlock. Instrumented runs maintained credit conservation, never ex-



**Figure 2:** PRESSURE-CAP3 under X-direction skew at  $L = 8$ , injection 0.10. The fully skewed saturated point is excluded from the latency curve.

**Table 1:** Tornado at  $L = 8$ , injection 0.9.

| Mode          | Thr./base | Latency | Lat./base |
|---------------|-----------|---------|-----------|
| PRIVATE-V4    | 1.000     | 49.37   | 1.000     |
| RTS           | 0.994     | 58.25   | 1.180     |
| Owner-RR      | 0.435     | 157.03  | 3.181     |
| PRESSURE-CAP3 | 1.000     | 46.97   | 0.952     |

ceeded the owner cap, and satisfied

$$N_{\text{alloc}} = N_{\text{owned}} + N_{\text{reclaim}}, \quad (4)$$

$$N_{\text{migration}} = N_{\text{reclaim}} + N_{\text{handoff}}. \quad (5)$$

Per-direction allocation totals matched aggregate counts, and observed cross-direction POOL read conflicts were zero. Under cap three, peak borrowing was one slot, confirming that the peer retained its final credit.

### 6.3 Persistent X-direction skew

The  $8 \times 4 \times 4$  X-biased workload sends each packet three minimal X hops and varies the probability of choosing +X. It preserves spatially distributed destinations while controlling directional imbalance. Figure 2 reports  $L = 8$ , injection 0.10, and four total VCs per input in the baseline.

At 50/50 and 60/40, TIDAL pays a 2.3% latency penalty because little cold-side capacity is stranded. The penalty falls to 1.6% at 70/30 and approximately breaks even at 80/20. At 90/10, latency falls by 9.2% while throughput rises by 2.1%. At 100% +X, the private baseline saturates, so a latency ratio is not a valid below-saturation comparison; the meaningful observation is 1.818 times delivered throughput during the measurement window.

Using completion/acceptance near one as a coarse criterion, the highest tested stable offered rates are 0.08 for PRIVATE-V4, 0.10 for cap three, and 0.12 for cap four. Cap three therefore extends the tested stable bracket by 25%; cap four provides an optimistic 50% upper bound. These are discrete brackets, not precise saturation thresholds.

### 6.4 Tornado and policy comparison

On  $4 \times 4 \times 4$ , torus tornado advances each packet by +1 in X, +1 in Y, and +1 in Z, making W, S, and D true inputs persistently hot. Table 1 shows  $L = 8$  and injection 0.9.

Pressure cap three lowers latency by 4.85% without changing throughput materially because an output bottleneck limits tornado first. It captures about 83% of the

latency benefit of cap four while preserving one peer credit. Only approximately 0.49% of POOL allocations use on-demand reclaim: stable ownership handles most demand, and reclaim resolves occasional mismatches. Owner round-robin loses 56.5% throughput because blind rotation can place credits at an inactive peer and has no demand-driven correction.

### 6.5 Where 1R1W loses

The phase schedule is a real bandwidth constraint. Against PRIVATE-V4, uniform traffic at  $L = 8$ , injection 0.9 raises latency by approximately 3.2%. X-opposite traffic raises latency by 3–15% because both members of a pair compete for one shared read. Tornado at  $L = 1$  raises latency by 25.1% because the credit loop is too short for additional capacity to repay the phase cost. Reversing the hot X direction every 16 cycles raises latency by 7.8% while ownership and old traffic drain in opposite directions.

VC scaling supports the same interpretation. Near the measured knee, a six-VC organization reaches a 1.27 throughput ratio and a 0.56 latency ratio, extending the stable bracket from 0.12 to 0.15. At eight VCs, where private capacity already hides more of the credit loop, the ratios shrink to 1.08 and 0.83, extending 0.18 to 0.20.

## 7 Hardware Interpretation

The evaluated comparison establishes equal buffer capacity and a one-read, one-write POOL service limit. This is a plausible manufacturing advantage over an ideal shared bank that must serve both directions simultaneously, but it is not yet an area or energy result. TIDAL adds owner bits, physical-slot mapping, selectors, arbitration, control wiring, and a potentially critical ownership update path. We therefore claim reduced shared-port demand, not zero hardware cost. RTL synthesis against private and dual-access pool implementations is required to quantify area, energy, and frequency.

## 8 Limitations and Next Steps

The present evidence is preliminary and simulator-level. First, demand reclaim updates ownership with zero additional cycles; a one-cycle and parameterized grant path should measure timing sensitivity. Second, finer 0.005 injection steps, longer drain-aware runs, and at least five seeds are needed around every saturation knee. Third, a directed simultaneous E/W-write microbenchmark should stress the write guard. Fourth, multi-flit packets require an explicit packet/VC ownership-retention rule. Finally, RTL synthesis should compare private buffers, 1R1W TIDAL, and an idealized dual-access pool.

## 9 Related Work

Virtual-channel flow control and its buffer/throughput tradeoffs were formalized by Dally [3]. Deadlock-free routing can be constructed through channel-dependency restrictions [4] or broader adaptive conditions [5]. TIDAL preserves a private deterministic escape resource rather than changing those guarantees. Garnet provides the de-

tailed router and credit model used here [1]; gem5 provides the surrounding simulation infrastructure [2]. Our contribution is specifically the combination of direction-paired equal-capacity storage, a structurally 1R1W shared bank, and demand-aware ownership credits.

## 10 Division of Labor

The project was completed with equal overall contribution: Changxun Pan 50% and Liming Wang 50%. Both contributors participated in architecture discussion, debugging, correctness review, experiment review, interpretation, writing, slide production, and presentation preparation.

**Liming Wang** proposed the original direction-paired pooling concept, implemented and evaluated the earlier 1R2W organization, expanded traffic and evaluation coverage, reviewed the 1R1W and credit-policy behavior, and co-analyzed and co-authored the final deliverables.

**Changxun Pan** co-developed the design and finalized its complementary 1R1W phase schedule, designed pressure, sticky-tie, cap-three, and reclaim control, implemented the final mechanism and instrumentation, ran and cross-checked major 1R1W experiments, and co-analyzed and co-authored the final deliverables.

## 11 Conclusion

TIDAL shows that buffer elasticity does not require an unconstrained global pool. Pairing opposite directions and preserving private escape capacity lets a four-slot adaptive pool move storage toward sustained pressure while retaining equal total capacity. A complementary schedule enforces one shared read and one shared write per cycle. The design is effective under persistent directional skew and long credit loops, but balanced demand, short links, and fast reversals expose its serialized-access cost. Pressure-aware credit movement is essential: blind ownership rotation is substantially worse. The next decisive step is to add grant delay and synthesize the control and storage organization so that performance gains can be judged against measured area, energy, and timing.

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